# Compliance sheet — star-moon

Everything a supplier qualification, a legal review or an export check usually asks about star-moon 0.1.0, in one document.

## 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-MOON-EN · v1.3 · 2026-10-16 |
| Package | star-moon |
| Version | 0.1.0 |
| Capability | star.orbit.moon |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_moon |

## 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

## 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

## 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_moon-0.1.0-
py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_moon-0.1.0-py3-none-any.whl | a50cb2e2e106001163cd963b32cfdf612904dbf06417c388f081dd2f520e222d |
| star_moon.py | ce41f15b68f4841d07880ec30babf3cb98a4e8c80bdb0ec867b6f6a0872cd769 |
| star_moon-0.1.0-source.zip | 8d4319952b295c78dc288a8e37acf8664d62df20eb4f6e29f261c588b489b582 |

# 5. What it does, and its limits

Geocentric Moon position for 1950-2050 from the Astronomical Almanac series; measured within 0.36 deg and 1200 km of JPL DE440 over 2002 dates, with the same envelope against astropy and PyEphem.

Not an ephemeris: 0.36 deg is about 2300 km at the Moon's distance. Good for eclipse and illumination geometry, third-body direction and visibility planning; not for navigation, occultation timing or tide-accurate work.

- `moon_vector_km(jd_day, jd_frac=0)`, `moon_ra_dec_deg`, `moon_distance_km` for Julian dates (TDB) from 1950-01-01 to 2050-01-01; outside, `ValueError`. Reproduces Vallado Example 5-3 within 0.4 m.
- Measured accuracy of the series (2002 dates, three ephemerides): direction within 0.36 deg of JPL DE440 (95 % of the dates within 0.20 deg), distance within 1200 km (288 km on average). astropy and PyEphem give the same envelope.
- Frame: mean equator and equinox of DATE, like `star_sun`; rotate before mixing with J2000/GCRF vectors.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a date outside 1950-2050.

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_moon_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-moon}: Where the Moon is, to a third of a degree},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_moon/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts